

TIVYAKU, Russian Federation

WMO: 317540

Lat: 48.600N

Lon: 137.050E

Elev: 403

StdP: 96.58

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.6	-35.4	-39.6	0.1	-37.3	-37.5	0.1	-35.2	6.9	-17.8	6.0	-17.1	0.0	230	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	29.0	20.3	27.1	19.6	25.3	18.7	22.0	26.2	21.0	25.0	20.0	23.8	1.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	16.2	23.9	19.7	15.1	23.0	18.6	14.1	22.0	66.7	26.1	63.0	24.9	59.4	23.7	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	5.4	4.9	-41.0	32.0	1.7	1.6	-42.2	33.2	-43.2	34.1	-44.2	35.0	-45.5	36.2		
(5)			-41.0	24.7	1.7	1.7	-42.3	25.9	-43.3	26.9	-44.2	27.9	-45.5	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-22.8	-19.2	-10.7	0.7	9.2	14.9	18.9	17.3	11.0	1.6	-10.9	-20.6
	DBStd	15.48	5.03	5.42	5.57	4.75	3.81	3.59	2.64	2.86	3.86	4.38	6.45	5.40
	HDD10.0	4691	1016	817	641	281	62	4	0	1	33	261	626	950
	HDD18.3	7053	1274	1050	899	530	284	113	26	53	222	518	876	1208
	CDD10.0	754	0	0	0	1	37	151	275	228	62	1	0	0
	CDD18.3	74	0	0	0	0	0	9	42	22	1	0	0	0
	CDH23.3	868	0	0	0	2	46	212	433	160	16	1	0	0
	CDH26.7	190	0	0	0	0	7	52	111	19	1	0	0	0
Wind	WSAvg	1.0	0.7	0.9	1.2	1.2	1.2	1.1	0.8	0.8	0.9	1.1	1.2	0.9
Precipitation	PrecAvg	817	19	19	36	45	70	92	144	161	91	54	46	30
	PrecMax	1166	36	57	143	89	162	174	328	291	240	120	99	68
	PrecMin	565	8	2	5	11	30	15	15	57	28	13	10	1
	PrecStd	160	9	11	29	20	36	44	79	60	47	28	20	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.6	-0.1	8.6	20.8	27.7	30.8	31.7	29.0	25.5	19.6	7.8	-1.5
		MCWB	-6.9	-2.4	2.4	10.2	15.7	20.2	22.3	21.6	18.4	11.9	3.9	-2.7
	2%	DB	-9.3	-3.5	4.8	16.2	24.4	28.0	29.6	26.8	22.9	14.7	4.1	-6.1
		MCWB	-10.2	-5.6	0.5	8.0	13.8	18.8	21.3	20.4	16.6	9.0	1.6	-7.0
	5%	DB	-11.8	-6.6	2.0	12.3	21.9	25.7	27.7	25.1	20.8	11.7	1.8	-9.4
		MCWB	-12.7	-8.0	-1.2	5.4	13.0	17.9	20.5	19.5	15.4	7.5	0.5	-10.1
	10%	DB	-14.1	-9.1	-0.2	9.3	19.1	23.5	25.8	23.6	18.4	9.3	-0.5	-12.0
		MCWB	-14.8	-10.5	-2.6	3.9	11.3	16.8	19.7	18.9	14.3	6.1	-1.7	-12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.7	-2.6	2.9	11.1	16.9	22.2	23.8	23.0	19.5	12.4	4.8	-2.7
		MCDB	-5.6	-0.2	7.3	19.7	25.0	27.1	28.6	27.0	24.0	17.7	6.9	-1.5
	2%	WB	-10.0	-5.7	0.9	8.3	15.1	20.2	22.3	21.7	17.7	10.1	2.1	-6.9
		MCDB	-9.2	-3.6	3.9	14.8	22.3	26.0	27.3	24.7	21.4	13.4	3.2	-6.1
	5%	WB	-12.6	-7.9	-0.9	6.2	13.6	18.8	21.3	20.8	16.3	8.2	0.3	-10.0
		MCDB	-11.8	-6.5	1.8	11.2	20.1	24.0	26.0	23.6	19.5	10.8	1.7	-9.3
	10%	WB	-14.8	-10.3	-2.5	4.4	12.2	17.6	20.4	19.7	14.9	6.5	-1.8	-12.6
		MCDB	-14.2	-9.1	-0.1	8.6	17.8	22.3	24.6	22.6	17.5	8.9	-0.5	-11.9
Mean Daily Temperature Range	5% DB	MDBR	14.1	16.8	16.3	13.1	14.8	14.3	12.4	11.0	13.0	12.0	10.8	11.8
		MCDBR	15.5	18.5	18.2	19.5	22.0	18.9	16.4	14.0	16.8	17.5	12.6	12.6
		MCWBR	14.4	16.3	14.2	11.9	12.3	10.7	8.7	7.8	10.3	11.5	10.3	11.7
	5% WB	MCDBR	15.2	17.5	15.7	17.5	18.5	15.9	13.5	11.1	13.8	15.3	11.5	12.0
		MCWBR	14.2	15.5	12.4	11.4	11.2	10.0	7.9	7.1	8.9	10.9	9.6	11.3
Clear-Sky Solar Irradiance	taub	0.282	0.330	0.398	0.505	0.539	0.494	0.479	0.429	0.396	0.414	0.365	0.294	
	taud	2.092	1.951	1.822	1.750	1.810	2.027	2.122	2.273	2.281	2.039	2.020	2.057	
	Ebn at Noon	754	786	786	741	736	778	784	811	799	682	623	670	
	Edh at Noon	87	128	174	208	204	166	148	122	110	118	95	79	
All-Sky Solar Radiation	RadAvg	1.83	3.02	4.35	4.55	5.15	5.52	5.21	4.45	3.57	2.32	1.48	1.35	
	RadStd	0.08	0.09	0.29	0.45	0.41	0.65	0.45	0.31	0.26	0.20	0.15	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page